Do investors value corporate social responsibility in emerging markets?

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Abstract

The paper examines the impact of corporate social responsibility (CSR) on firm value for a large sample of Indian firms, arguably an emerging market with poor investor protections and legal enforcements. We proxy CSR by Bloomberg score on the extent of a company's Environmental, Social, and Governance disclosures, and find positive valuation effects associated with CSR performance. We further find that CSR performance is more valuable to standalone firms compared to business group firms. We show that the positive valuation effects associated with CSR is attributed by lower cost of fund, and higher operating cash flow.

Keywords: corporate social responsibility; stakeholder theory; firm value; cost of fund; business group; emerging markets; India

JEL classification: G32; G34; M14.

Introduction

In the recent decades, firms are dedicating substantial resources in corporate social responsibility (CSR) activities. Regulators of various countries also mandate firms to allocate a portion of it profits in CSR activities. Indian regulators[1], for example, require a firm to allocate 2 percent of its average profits of the prior three years in CSR activities. The rise of CSR investments has subsequently engrossed academic and business press[2] to examine the cost-benefits of CSR investments. Despite the growing debate on the cost-benefits of CSR investments around the world including emerging markets, most of studies emphasize on U.S data (Liang & Renneboog, 2017) and find mixed results[3]. Although handful studies also consider firms from various countries, these studies are likely to be inadequate to describe cross-sectional differences in the consequence of CSR investments on firms within countries**.** Liang and Renneboog (2017) also argue that a firm's explicit and implicit contractual legal enforcements, which are likely to be derived by the legal and enforcement mechanisms of the domicile country, determine the impact of CSR**.** Motivated by this insight, we use a large sample of Indian firms, arguably an emerging market with poor investor protections and legal enforcements[4], to shed light on this debate. Particularly, we propose two opposing hypotheses concerning CSR performance; the stakeholder value maximization and the shareholder value maximization.

The stakeholder value maximization hypothesis exhibits a positive effect of CSR performance on the shareholder value because focusing on the welfares of other stakeholders rises their inclination to support's firm operations, thereby enhancing firm performance and firm

[1] For further information, Please refer the Indian Company Act, 2013, http://www.mca.gov.in/Ministry/pdf/CompaniesAct2013.pdf.
[2] http://www.livemint.com/Opinion/1wIQwFPRyRckBMg5IugW1K/Why-the-CSR-law-is-not-a-success.html.
[3] Please refer Malik (2015) detail literature review.
[4] See, La Porta et al. (1999)

value. This view is populated from the contract theory of firm survival (Alchian & Demsetz, 1972; Coase, 1937; Hill & Jones, 1992; Jensen & Meckling, 1976). The contract theory postulates that the firm is a nexus of contracts between shareholders and other stakeholders wherein stakeholders are supplier of resources required to operate the firm in exchange for claims outlined either with explicit contracts, (e.g., wage contracts or product warranties) or with implicit contracts (e.g., promise of job security). Unlike explicit contracts, implicit contracts are vague and have limited legal supports. Firms, therefore, can violate implicit contracts without legal recourse from other stakeholders. Nevertheless, firms' commitment to endorse implicit contracts can increase the value of implicit contracts (Cornell & Shapiro, 1987). Firms, consequently, can use CSR to build reputation for endorsing implicit contracts. This is particularly favorable to Indian firms where the legal enforcements are limited and firms' founders with a complex ownership[5] are not only decide how the firm would be operated , but also how the profits are shared among shareholders (Bertrand, Mehta, & Mullainathan, 2002; Claessens, Djankov, Fan, & Lang, 2002; Claessens, Djankov, & Lang, 2000; Claessens & Yurtoglu, 2013; La Porta, Lopez-de-Silanes, & Shleifer, 1999). This causes conflicts between large and minority shareholders. These firms, as a result, are vulnerable to lower valuation because of the potential expropriation risk (Mahajan, 1990; Villalonga & Amit, 2006). Founders of these firms, therefore, may use CSR investments, as an alternative commitment mechanism, to convey a signal to minority shareholders that their interests would not be expropriated. Overall, the above arguments suggest that the interests of shareholders, particularly founders, and other stakeholders, including minority shareholders, are more aligned in firms showing greater social responsibility. Other stakeholders, therefore, are likely to contribute to firms' long-term

[5] Cross-holding ownership and pyramidal ownership, see, Anderson and Reeb (2003); Bertrand et al., (2002); Claessens and Fan (2002).

profitability and efficiency (Freeman, Wicks, & Parmar, 2004; Jawahar & McLaughlin, 2001; Jensen, 2001). Likewise, minority shareholders are less likely to discount stock prices, because of expropriation risk.

The shareholder value maximization, on the other hand, hypothesizes that managers engage firms in social responsible activities in order to help other stakeholders at the price of shareholders (Pagano & Volpin, 2005; Surroca & Tribo, 2008). Aupperle et al. (1985) advocate that social responsible firms may rise competitive disadvantage, because these firms incur costs that might otherwise avoided. For instance, acquiring environmentally friendly equipment and executing stricter quality controls. Unveiling social responsibility information also include costs, such as data collection, communication and auditing (Branco & Rodrigues, 2006). Moreover, firms' insiders, such as founders, may have an incentive to increase CSR expenditure in order to attain a high CSR rating, mainly when founders control firms' resources. This is because a favorable CSR rating can advance their status[6] as individuals who care about society, employees, and the environment (Barnea & Rubin, 2010). Socially responsible firms, as a result, are likely to employ too much resources in nonproductive CSR activities, thereby lowing profitability and firm value.

To examine hypotheses empirically, we quantify the intensity of CSR performance by Bloomberg's proprietary score based on the extent of a company's Environmental, Social, and Governance (ESG) disclosures. By using data period over 2007 to 2016, we report that CSR performance has a significant positive effect on firm value, measured by *Tobin Q*. Our results are robust with various control variables, including firm-fixed effects. This effect is also economically significant; *Tobin Q* rises by 13.75 percent relative to an average of *Tobin Q* as

[6] Videras and Owen (2006) use a broad multi-country sample, and show that individuals who contribute the public good of environmental protection attain higher level of life satisfaction.

CSR variable progresses from the first to the third quantile of the distribution. The result highlights the significance of firm investing in non-financial capital, for example social capital. Furthermore, we provide evidence that there is a significant heterogeneity in the valuation effects between business group firms and standalone firms. Business group firms are always criticized for their low transparency concerning corporate decision-making (Khanna & Palepu, 1999; Khanna & Yafeh, 2007). Furthermore, complex intra-group transactions make challenging for outsiders to comprehend the motive of founders (Dewenter, Novaes, & Pettway, 2001). Consistent with this view, we find that the positive effect of CSR performance is less prevailing for business group firms compared to standalone firms.

We next address the identification issue, because our aforementioned relationship can be vulnerable to reverse causality. We use the two stage least square (2SLS) regression method to examine the valuation effect of CSR. Particularly, we control endogeneity problem by using instrument variables, industry-specific CSR, an indicator of firm's past profitability, and a firm's commitment to resolve investors' complains, that are expected to be related to a firm-level CSR, rather than firm value (Cheng, Ioannou, & Serafeim, 2014; El Ghoul, Guedhami, Kwok, & Mishra, 2011). We again confirm our aforesaid relationship.

We next explore the channels through which CSR performance contributes to firm value. We, particularly, study the influence of CSR performance on the cost of capital and cash flows, determinants of firm value (Damodarn, 2012). The effect of CSR on the cost of capital is motivated by the following two reasons. First, previous studies document that socially worried investors are less likely to include low CSR firms in their portfolios (Heinkel, Kraus, & Zechner, 2001; Kacperczyk, 2009). Decreasing the relative size of a firm's investors' base will subsequently increase cost of capital of low CSR firms, thereby reducing firm value (Merton, 1987). Second, related studies suggest that socially responsible firms are less likely to engage in earnings managements (Hong & Andersen, 2011; Kim, Park, & Wier, 2012), thereby improving corporate transparency. We postulate that the extent to which CSR performance improves corporate transparency, there would be less information asymmetry between insiders and outsiders, leading to lower cost of capital (Hail & Leuz, 2006). We report that firms showing greater social responsibility observe lower cost of capital, particularly cost of equity. We also demonstrate that the negative effect of CSR on cost of capital is less prevailing for business group firms compared to similar standalone firms.

In the second channel, we explore the impact of CSR performance on operating cash flows and dividend payouts. Cheng et al. (2014) document that socially responsible firms experience less capital constraints. To the extend CSR performance facilitates firms to alleviate capital constraints, firms are anticipated to invest in those positive NPV projects yet they are not pursued because of capital constraints, thereby enhancing future operating cash flows of firms. Dhaliwal et al. (p 2011, p. 62) also notice, "...some CSR projects have direct implications for positive cash flow even in the near future. For example, practices related to protecting the environment and improving employee welfare can reduce potential litigation and pollution cleaning costs, boost employee morale." The empirical result confirms that firm with superior CSR performance have greater future cash flows. Consistent with Whited & Wu (2006) that dividend paying firms face less financial constraints, we find that socially responsible firms pay more cash dividends.

Overall, our study produces empirical findings that advocate that investors assign more valuation to socially responsible firms. From a firm's prospective, our results indicate that the benefits of CSR investments outweigh the cost of these activities. Our paper, therefore, contributes to the ongoing debate on impact of CSR performance on firm value in several ways. First, given the growing significance of CSR in emerging markets, our study initiates and examines the impact of CSR performance on firm value for an emerging market. Our study, therefore, extends this debate in a novel institutional setting in order to provide robustness to previous empirical findings. Second, emerging markets are dominated by business group firms that, by construction, are less transparent compared to similar standalone firms because of intra-group transactions. We contribute to this literature by testing whether the impact of CSR significantly varies between business group firms and standalone firms. Our results highlight the role of organization structure in the relationship between CSR and firm value. Finally, we reconfirm the empirical findings of El Ghoul et al. (2011) by documenting that CSR performance facilitates firms to reduce cost of capital.

The rest of the paper is organized as follows: Section 2 describes data and methodology. Section 3 presents empirical results. Section 4 concludes the study.

2. Data and Methodology

We obtain data for this study from two sources. We collect financial data from the Prowess, a database marinated by the Centre for Monitoring Indian Economy (CMIE). The Prowess database is generally used by earlier research on Indian firms (Bertrand et al., 2002; Gopalan, Nanda, & Seru, 2007; Khanna & Palepu, 1999).

Previous studies note that CSR is a multidimensional construct, and thereby, a signal variable, such as employee satisfaction, cannot obverse the complete picture concerning the performance of CSR investments. We, therefore, use proprietary Bloomberg score based on the extent of a company's Environmental, Social, and Governance (ESG) disclosure. Bloomberg score varies from 0.1 for firms that disclose a minimum amount of ESG data to 100 for those that disclose every data point collected by Bloomberg. Thereafter, each data point is weighted in terms of importance, with data such as Greenhouse Gas Emissions carrying greater weight than other disclosures. The score is also tailored to different industry sectors. In this way, each firm is only evaluated in terms of the data that is relevant to its industry sector.

The Bloomberg provides data from 2007 onwards and therefore, our study covers data period over 2007 to 2016. We exclude firm-year observations with missing data and negative sales. We also exclude financial and government firms from our sample. Our final sample covers 3837 firm-year observations for 630 firms. We further winsorize all variables at the bottom and the top one percent in order to alleviate the effect of outliers.

2.1. Variable formation

We measure firm value by *Tobin Q*. *Tobin Q* is a market-based measure and hence considers investors' forward-looking valuation. Tobin *Q* also reveals the value of intangible assets better. *Tobin Q* is the market value of assets divided by book value of assets. The market value of assets is calculated by market value of equity plus book value of debt. We calculate the cost of capital by weighed average cost of capital. Cost of equity is calculated by the Capital Assets Pricing Model. To do so, the beta of equity is measured by rolling window of 24 months. Risk free rate is the default value for the risk-free rate is the country's long-term bond rate (ten year bond). We collect the cost of debt from the Bloomberg. Operating cash flow (*Operating cash flow*) is calculated by net operating profit after tax, divided by total assets. Dividend payout (*Dividend payout*) is dividend amount divided by profit after tax.

We also include control variables. We control for firms' affiliation with business groups by a dummy of business group (*Group*) that takes a value 1 for firms affiliated with a business group and 0 otherwise. We control for firm size, calculated as natural log of total assets (*Size*), and for leverage (*Lev*) using a ratio of total borrowing to total assets. Dividend paying stocks are controlled by dividend yield dummy (*Dividend dummy*) that takes a value 1 if a firm has paid dividend in the fiscal year and 0 otherwise. Firm age (*ln (Firm Age))* is the nature logarithm of year since firm incorporation. Board independence *Board Ind (%)*) is measured by the percentage of independent directors in a corporate board.

3. Empirical results

3.1. Descriptive analysis

The descriptive statistics of variables used in the analyses are provided in Table 2. Table 1 reports the details concerning formation of the variables. For our sample firms, the mean (median) value of *Tobin Q* is 1.32 (0.83), with the interquartile (Q3-Q1) range of 0.93. The mean (median) value of CSR performance is 14.95 (13.22), with the interquartile range of 4.20. The average sample firm's cost of capital is 9.89 percent, with 10.99 percent cost of equity, and 6.74 percent cost of debt. Our sample firm pays 29.05 percent of total profit as dividend, with average age of about 29 years ($e^{3.36}$). Our sample firms have an average *Lev* of 33 percent. The average size (*Size*) of sample firms is around Rs. 7,943 million ($e^{8.98}$), with median firm size around Rs. 9,509 million ($e^{9.16}$), indicating that our sample firms are skewed to large firms.

Table 3 reports the Pearson correlation coefficients between variables used for the study. At first glance, it can be observed that CSR performance is positive and significantly correlated with a year ahead *Tobin Q*, and negative and significantly correlated with cost of equity, including cost of debt. This suggests that socially responsible firms perform well, with low cost of fund. As expected, firm CSR is highly correlated with firm size (*Size*) and *Tobin Q*, giving a reason to control firm size to obtain the impact of CSR performance on *Tobin Q*. Several of the other correlations are significantly different from zero displaying the significance of containing these variables as control in our main empirical analysis. Nevertheless, none of the correlations are extremely high, representing that multicollinearity will not be material in our regression analysis.

3.2. Multivariate analysis

3.2.1. The effect of CSR on firm performance

In this subsection, we examine the impact of CSR performance on firm value. To do so, we follow the following regression equation:

$$Tobin\ Q_{it+1} = \propto + \beta_1 CSR + Controlvariables + \varepsilon_{it} \qquad (1)$$

Here, firm and year are represented by *i* and *t,* respectively. *Tobin Q* is a proxy of firm value. *CSR* measures the performance of CSR investments concerning environment, social and governance aspects. We include all control variables described in Section 2.2. We also include industry fixed effects to control for invariant industry effects and year fixed effects to control for time trends in firm value. We follow the two-digit National Industry Classification (*NIC*) to classify industry. To account for time series and cross-sectional correlations in our regressions, we follow Petersen (2009) and estimate cluster standard errors at firm and year levels.

Table 4 presents the empirical results. In Column 1, the coefficient of *CSR* is positive and statistically significant at 1 percent level. This implies that investors pay premium for firms showing social responsibility. The degree of the effect is also economically significant, since firm value (*Tobin Q*) rises by 13.75 percent as *CSR* variable progresses from the first to the third quantile of the distribution. To compute this effect, we first multiply the interquartile range of *CSR* variable, from Table 2, with the coefficient of *CSR* from Column (1) of Table 4. This calculation yields (0.021*8.64) a measure of increased *Tobin Q* when *CSR* increases from the first to the third quartile of the distribution. We next compare this increase in *Tobin Q* with the average (1.32) of *Tobin Q*. This comparison reveals that an increase in *CSR* from the first to the third quartile of the distribution results in 13.75 percent (0.18/1.32) increase in firm value (*Tobin*

Q) relative to the sample average of *Tobin Q*. The comparison, alternatively, may be calculated with comparison of the interquartile range of *Tobin Q.* An increase in CSR from the first to third quartile leads to 19.3 percent increase in firm value relative to the interquartile range of firm performance (*Tobin Q*, 0.18/0.93).

Recent evidence reveals that better governed firms are likely to have greater valuation, and simultaneously may be correlated with CSR activities (Chauhan, Dey, & Jha, 2016; Jain & Jamali, 2016). In such case, it is likely that *CSR* variable is merely proxying firm-level governance, leading to omitted variable bias. To address this caveat, we control firm-level governance by board independence (the proportion of board consisting of independent directors). Column (2) shows that the coefficient of *Board Ind (%)* is negative and significant at 10 percent level. We again confirm the positive effect of *CSR* on firm value. This evidence indicates that our *CSR* variable is related with firm value and is not merely showing up as the consequence of firm-level governance.

Our results could be vulnerable to time-invariant omitted variable biasness. For instance, a high quality manager is more likely to be associated with higher firm value. At the same time, the manager can endorse *CSR* activities, without having any significant benefits. In such cases, a high quality manager, as an omitted variable, can institute our suggested relationship. To alleviate this anxiety, we rerun the regression equation (1) by substituting firm-fixed effect with industry-fixed effects. In Column (3), we find consistent results as reported in Column (2). In Column (4), we control the persistence of firm value by using value of *Tobin* Q_t and find consistent results.

In Column (5), we examine the marginal effects of *CSR* for firms affiliated with business group. To do so, we include an interaction of *CSR* and *Group* in regression equation (1). The coefficient of *CSR* and *Group* is positive and significant at 1 percent level, signifying the marginal effect of individual variables on firm value. The coefficient of *CSR*Group* is negative and statistically significant at 5 percent level. The value of coefficient suggests that the positive effect of *CSR* for standalone firms (0.042) is almost three times compared to business group firms (0.042+-0.029=0.013), indicating about the variability of CSR's effect on firm value with respect to firm affiliation.

Our results concerning control variables are consistent with previous studies. For instance, the positive coefficient of *Dividend dummy* reveals that investors pay premium to divided paying firms.

Overall, our empirical results reveal that investors pay premium to socially responsible firms. We, therefore, prove the *stakeholder value maximization hypothesis*. That is, the consideration of other stakeholders' concern provides valuable resources to the firm, leading to greater firm value. Our regression models explain approximately 23 percent to 85 percent of variation of firm value, indicating the statistical power of our regression models.

3.2.2. Robustness analysis

In the corporate finance studies, there is always a possibility of reverse causality. Our concern is mainly motivated by Waddock and Graves's (1997) two alternative hypotheses associated with CSR-firm performance relationship. *Good management hypothesis* proposes that better socially responsible firms perform well, since CSR performance improves firm relationship with stakeholders. *The slack resource hypothesis*, on the other hand, postulates that well performed firms have more resources to spend on CSR activities, thereby attaining higher standard. We address this concern by the two stage least square regression with appropriate instrument variables.

To do so, we consider three instrumental variables. First, Following Cheng et al (2014) and El Ghoul et al., (2011), we consider the mean value of *CSR* for each industry, excluding the firm itself. The intuition is that the CSR activities of a firm is likely to be correlated with other firms within industry (Ioannou & Serafeim, 2012). By excluding firm itself, we allow instrument variable to vary across firms, even within the same industry. Second, following Jiao (2010), we consider an indicator of negative earnings in the previous year, because an indicator is dummy variable and therefore is not likely to capture the level of expected cash flow in the future. Jiao (2010) argues that CSR investments is related to historical earnings. Third, we deploy the percentage of investors' complained resolved in a given year. That is, the number of complains resolved in a given year, divided by the number of complains at the beginning of the year plus the number of complaints received during the year. We expect that the intensity of resolving complains would be associated with socially responsible firms, because socially responsible firms are more likely to resolve complains in given year in order to sustain corporate image. On the other hand, there is no rationale or empirical evidence that advocates that investors' complains are correlated with firm value. We also include a lagged variable of *Tobin Q* as an independent variable in the second stage regression. We then use GMM technique developed by Blundell and Bond(1998) to estimate dynamic panel model.

Column (1) and (2) of Table 5 report the results from the first stage and the second stage regression, respectively. We also perform postestimation tests. We perform overindentifcation test, Hansen's J statistic (Hansen, 1982). The p-value (0.547) of this test reveals that instruments satisfy the condition of exogeneity and is appropriate as instruments. In the first stage, the coefficients of *Ind CSR, Loss dummy,* and *%Complain resolved* are correlated with a high value of t-statistic. Again confirms the relevance of our instrument variables. In the second stage, the coefficient of *CSR* is positive and statistically significant at 1 percent level, with almost same magnitude as reported in Column (1) of Table 4.

Overall, these results suggest that the endogeneity concerns are not likely to govern our main finding.

3.2.3. The channels through which CSR performance improves firm value

In this subsection, we provide the empirical results for the possible channels by which *CSR* improves firm value.

We first report results concerning the cost of capital, cost of equity, and cost of debt. To do so, we use the following regression model.

$$Cost\ of\ fund_{it+1} = \propto + \beta_1 CSR + Control variables + \varepsilon_{it} \quad (2)$$

Here, firm and year are represented by *i* and *t,* respectively. *Cost of fund* considers a variable constructed out of cost of capital, cost of equity, and cost of debt. *CSR* measures the performance of CSR investments concerning environment, social and governance aspects. We include all control variables described in Section 4.2. We also include industry fixed effects to control for invariant industry effects and year fixed effects to control for time trends in cash holdings. We follow the two-digit National Industry Classification (*NIC*) to define industry. To account for time series and cross-sectional correlations in our regressions, we follow Petersen (2009) and estimate cluster standard errors at firm and year levels.

Table 6 reports the results. Column (1), (2), and (3) of Panel A report results for cost of capital, cost of debt, and cost of equity, respectively. We find that the coefficient of *CSR* is negative, but insignificant for cost of debt. This implies that firms showing better social responsibility have significantly lower cost of fund, particularly cost of equity. Economically, the cost of equity reduces by 1.18 percent relative to mean value of cost of equity as *CSR* variable moves from the first to the third quantile of the distribution[7]. We interpret this results as important channel through which the market prices socially responsibly firms.

Panel B examines the marginal effect of *CSR* for business group firms. As expected, the coefficient of *CSR**Group* is positive, but significant only for cost of equity. This suggests that business group firm showing greater social responsibility have more cost of equity compared to similar standalone firms.

We next explore the impact of CSR performance on cash flows, particularly operating cash flow and dividend payout. To do so, we use the following regression equation:

$$Cah\ flow_{it+1} = \propto + \beta_1 CSR + Controlvariables + \varepsilon_{it} \quad (3)$$

Here, firm and year are represented by *i* and *t,* respectively. *Cash flow* considers a variable out of operating cash flow and dividend payout. We take in all control variables described in Section 4.2. We also include industry fixed effects to control for invariant industry effects and year fixed effects to control for time trends in cash holdings. We follow the two-digit National Industry Classification (*NIC*) to define industry. To account for time series and cross-sectional correlations in our regressions, we follow Petersen (2009) and estimate cluster standard errors at firm and year levels.

Table 7 presents the results. Panel A and Panel B report results for operating cash flow and dividend payout, respectively. In Panel A, the coefficient of *CSR* is positive and significant at 10 percent, demonstrating that socially responsible firms have greater future operating cash

[7] Please refer the economic significance of *Tobin Q* in order to infer calculation.

flow. Economically, operating cash flow increases by 4.16 percent relative to mean value of operating cash flow as *CSR* variable moves from the first to the third quantile of the distribution[8] The coefficient of the interaction variable (*CSR*Group*) also confirms that CSR performance is less important to business group firms as compared to standalone firms. This is consistent with our previous findings. Panel B reports also supports our results by showing that socially responsible firms pay more cash dividend.

Overall, this subsection provides empirical results associated with the channels through which CSR performance improves firm value. Particularly, we document that CSR performance is priced, and is associated with lower capital financing and therefore, is supporting firms to invest in more positive NPV projects, which in turn increasing future operating cash flow.

3.2.4. Operating performance and CSR performance

So far, we document that CSR performance increases firm value. We now examine the impact of CSR performance on future operating performance. To do so, we consider return on assets (*ROA*), profit before the interest, depreciation, and taxes, divided by total assets. In equation (1), we replace *Tobin Q* by *ROA*[9] of $year_{t+1.}$ Table 8 reports the results which are consistent with Table 5. The coefficient of *CSR* is positively significant, whereas the interaction of *CSR*Group* is negatively significant. Overall, Table 8 shows that our results are robust to alternative of firm welfares, i.e. operating performance.

3.2.5. CSR and firm value: Financial crisis

Thus far, we prove that CSR performance, on average, increases firm value. We next explore the period in which firm value is, on average, lower because of external negative shocks. We argue that if CSR investments facilitate firms to build inimitable intangible assets (Branco &

[8] Please refer the economic significance of *Tobin Q* in order to infer calculation.
[9] We also use returns on sales and returns on equity, and consistent results.

Rodrigues, 2006), it should compensate when overall market valuation is lower. To do so, we employ the 2008-2009 financial crisis, a period during which the overall valuation of firms declined surprisingly. Moreover, the advantage of concentrating on the financial crisis period is that it permits us to observe the impact of CSR performance on firm value unequivocally, because we consider CSR performance instantaneously before the external shock to explicate the change in firm value. We, therefore, can also eliminate spurious causality caused by the endogeneity problems.

To test whether our empirical results are also robust to a period of financial crisis, we consider firm value (*Tobin Q*) for the period of 2008-2009 and regress on *CSR* variable, measured at the end of 2007, including other control variables described in Section 4.1. Table 9 reports the results and confirm that investors place valuation premium on firms that are showing socially responsibility, even when the overall market valuation is low. We further show that investors explicitly demand lower cost of equity in order to show their trust on these firms, even in a financial crisis period. Economically, *Tobin Q* increases by 7.9 percent relative to mean value of *Tobin Q* (1.16) during the financial crisis period as *CSR* variable moves from the first to the third quantile of the distribution[10] (5.37), measured at the end of 2007. Nevertheless, the economic impact during the financial period is lower than the overall sample period (17.08 percent[11]). Overall, our results document that socially responsible firms suffer less during financial crisis.

4. Conclusion

The paper examines the impact of CSR performance on firm value for Indian firms, arguably an emerging market governing by poor legal enforcements and concentrated founder

[10] Please refer the economic significance of *Tobin Q* in order to infer calculation.
[11] Please refer the estimated coefficient of *CSR* variable from Column (3) of Table 4; that is 0.042.

ownership. Indian market, therefore, allows to explore the aforementioned impact in different market structure in order to provide further empirical evidences to contribute in the debate on whether CSR investments are value-enhancing, destroying, or neutral (Jiao, 2010). Using data from Bloomberg's proprietary score based on the extent of a company's Environmental, Social, and Governance (ESG) disclosures over the period of 2007-2016, we document a positive valuation effects associated with CSR performance. Furthermore, the valuation effects vary with firm affiliations; business group firms and standalone firms. We, mainly, find that the valuation effects are weaker for business group firms compared to similar standalone firms. Our result again highlights the significance of our study on Indian market. To the best our knowledge, we are the first to report such finding. We next examine the channel through which CSR performance impacts firm value. We find that socially responsible firms are having lower cost of fund and higher operating profits. These firms also pay more cash dividends.

The findings in this paper also have practical implications. Indian regulators enforce Indian firms to spend 2 percent of its average net profit over last three year in CSR activities. Our results should encourage managers to invest more in CSR activities, because CSR investments not only facilitate the wellbeing of society, but they also benefit the firm by improving firm value, the ultimate objective of mangers. Note that our results do not highlight the benefit-costs of regulators' enforcement in CSR spending. Recently, Manchiarju and Rajgopal (2017) document a negative valuation effects on the aforementioned regulation. We trust that this valuation effect may be a consequence of further taxing on firms because of compelling firms to spend on CSR activities, rather than being a volunteer firm to spend on these activities. Perhaps, regulators requisite to accept that if CSR investments are valuable to firms, they are likely to invest in CSR activities.

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Table 1: Description of variables

The table presents the definition of variables used in this study

Variable	Definition
Tobin Q	Total outstanding stock multiplied by the year-end stock prices plus total borrowing, divided by total assets.
Return on Assets (ROA)	Ratio of operating profits to total assets.
Cost of equity	We follow CAPM to calculate cost of Equity
Cost of debt	We follow Bloomberg's methodology to calculate cost of debt.
Operating cash flow (*CFO\TA*)	by net operating profit after tax, divided by total assets
Dividend payout (*Dividend payout (%)*)	Dividend amount divided by profit after tax.
CSR performance	Bloomberg score on the extent of a company's Environmental, Social, and Governance disclosures
Leverage(Lev)	Ratio of book value of total borrowings to book value of total assets.
Business Group (Group)	Dummy variable that takes a value 1 for firms that belong to a business group and 0 for standalone firms
Firm size(Size)	A natural log of total assets
Firm age (*ln (Firm Age))*	The nature logarithm of year since firm incorporation.
Dividend dummy	A dummy variable takes a value 1 for firms with dividend, and 0 otherwise
Board Independence (*Board Ind (%))*	The percentage of independent directors in a corporate board
Industry CSR *(Ind CSR)*	The mean value of *CSR index* for each industry, excluding the firm itself
Loss dummy	An indicator of negative earnings in the previous year
%Complain resolved	The number of complains resolved in a given year, divided by the number of complains at the beginning of the year plus the number of complaints received during the year.

Table 2: Descriptive statistics

This table presents distribution of variables for the sample firms. Table 1 reports the definition of variables.

Variable	Mean	Median	Std Dev	Q3	Q1
Tobin Q	1.32	0.83	1.31	1.52	0.59
Cost of Equity	10.99	10.98	1.90	12.13	10.00
Cost of Capital	9.89	10.16	1.87	11.09	8.93
Cost of Debt	6.74	7.40	3.41	9.02	5.34
CFO\TA	0.07	0.07	0.12	0.12	0.02
Dividend payout (%)	29.05	14.46	137.7	30.74	6.16
Size	8.98	9.16	1.98	10.36	7.64
Leverage	0.33	0.30	0.33	0.44	0.14
CSR	14.95	13.22	7.89	16.53	10.33
Board Ind (%)	52.47	50.00	12.30	60.00	45.45
ln(Firm Age)	3.36	3.33	0.63	3.78	3.00

Table 3: Correlation matrix

This table presents correlation matrix of all variables for the sample firms. Table 1 reports the definition of variables. The bold correlations are significant at 5% level.

	Tobin Q	*Cost of Equity*	*Cost of Capital*	*Cost of Debt*	*CFO\TA*	*Size*	*Leverage*	*CSR*	*Board Ind (%)*	*ln(Firm Age)*	*Dividend payout*
Tobin Q	1										
Cost of Equity	**-0.08**	1									
Cost of Capital	**0.15**	**0.66**	1								
Cost of Debt	0.02	**0.19**	**0.18**	1							
CFO\TA	**0.13**	0.00	**0.04**	**-0.09**	1						
Size	**0.19**	**0.31**	**0.21**	**0.23**	**0.06**	1					
Leverage	**-0.05**	**-0.05**	**-0.14**	**0.03**	**-0.13**	**-0.14**	1				
CSR	**0.11**	**-0.04**	0.02	**-0.04**	**0.11**	**0.34**	**-0.05**	1			
Board Ind(%)	**-0.04**	0.00	-0.02	**0.09**	**0.04**	**-0.03**	**0.05**	0.01	1		
ln(Firm Age)	0.00	0.02	**0.05**	-0.02	**0.04**	**0.08**	-0.01	**0.17**	**0.06**	1	
Dividend payout	0.00	0.01	-0.01	-0.00	-0.00	**0.04**	-0.00	**0.06**	**0.00**	**0.03**	1

Table 4: Corporate social responsibility and firm performance

The table reports the relationships between corporate social responsibility and firm performance. Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

Variable					
Intercept	2.074***	2.375***	2.266*	0.239	1.974
	(3.360)	(3.870)	(1.730)	0.300	(1.510)
CSR	0.021***	0.021***	0.019***	0.013***	0.042***
	(2.780)	(2.680)	(3.110)	(3.100)	(3.860)
*CSR *Group*					-0.029***
					(-2.250)
Group	0.020	0.036	0.172	-0.157	0.663***
	(0.230)	(0.420)	(0.700)	(-0.930)	(2.200)
Size	0.023	0.036	-0.075	0.038	-0.069
	(0.810)	(1.210)	(-1.030)	(0.670)	(-0.930)
Leverage	0.271	0.259	0.066	-0.189*	0.042
	(1.060)	(0.880)	(0.700)	(-1.820)	(0.430)
ln(Firm Age)	-0.006	-0.044	-0.277	-0.085	-0.289
	(-0.090)	(-0.620)	(-0.770)	(-0.450)	(-0.810)
Dividend Dummy	0.323***	0.327***	0.130***	0.084***	-0.001
	(4.950)	(4.670)	(2.650)	(2.470)	(-0.360)
Board Ind (%)		-0.004*	-0.001	0.000	-0.001
		(-1.690)	(-0.400)	(-0.040)	(-0.360)
Tobin Q				0.427***	
				(8.830)	
Industry Fixed Effects	Yes	Yes	No	No	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	No	No	Yes	Yes	No
Adjusted R^2	0.23	0.23	0.82	0.85	0.23
N	3837	3837	3837	3837	3837

Table 5: Corporate social responsibility and firm performance: 2SLS

The table reports the relationships between corporate social responsibility and firm performance. Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

Variable	**First Stage**	**Second Stage**
Intercept	-10.101***	-0.097
	(-4.120)	(-0.910)
Ind CSR	1.0236***	
	(26.19)	
Loss dummy	0.785***	
	(2.46)	
%Complain resolved	0.541**	
	(2.10)	
CSR		0.013***
		(3.190)
Size	1.029***	-0.006
	(6.910)	(-0.770)
Group	0.516	-0.023
	(1.540)	(-0.870)
Leverage	-0.005	-0.149***
	(-0.020)	(-3.40)
ln(Firm Age)	1.018***	0.043**
	(2.840)	(1.990)
Dividend Dummy	0.044	0.039
	(0.160)	(1.300)
Board Ind(%)	-0.004	0.001
	(-0.280)	(0.920)
Tobin Q		0.887***
		(38.300)
Adjusted R^2	0.260	0.701
Industry Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
N	3837	3837
Hansen J Statistic		1.21
Overindentifcation test		0.547

Table 6: Corporate social responsibility and cost of fund

The table reports the relationships between corporate social responsibility and cost of fund. Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

	Panel A			Panel b		
Variable	**Cost of Capital**	**Cost of Debt**	**Cost of Equity**	**Cost of Capital**	**Cost of Debt**	**Cost of Equity**
Intercept	11.109***	2.854***	11.201***	11.301***	2.633***	11.302***
	(36.89)	(4.74)	(42.81)	(39.16)	(4.03)	(35.72)
CSR	-0.011**	-0.012	-0.015***	-0.022***	0.003	-0.023***
	(-2.11)	(-1.45)	(-4.22)	(-3.08)	(0.16)	(-3.26)
CSR *Group				0.008	-0.017	0.015*
				(0.98)	(-0.88)	(1.74)
Group	0.181***	-0.005	0.176***	0.07	0.229	-0.024
	(2.22)	(-0.03)	(2.51)	(0.54)	(0.74)	(-0.16)
Size	0.058***	0.49***	0.283***	0.282***	0.491***	0.057***
	(2.33)	(10.34)	(12.93)	(12.92)	(10.35)	(2.31)
Leverage	-0.447***	0.996***	0.071	0.069	0.999***	-0.45***
	(-2.75)	(2.77)	(0.64)	(0.63)	(2.77)	(-2.75)
ln(Firm Age)	0.066	-0.143	-0.072	-0.072	-0.142	0.065
	(1.16)	(-1.28)	(-1.49)	(-1.49)	(-1.28)	(1.14)
Board Ind(%)	-0.001	0.011***	0.001	0.001	0.012***	-0.001
	(-0.36)	(2.35)	(0.27)	(0.22)	(2.39)	(-0.44)
Tobin Q	0.141***	-0.089	-0.13***	-0.129***	-0.09	0.142***
	(5.19)	(-1.42)	(-5.46)	(-5.42)	(-1.43)	(5.17)
Adjusted R^2	0.29	0.16	0.61	0.16	0.29	0.61
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
N	3837	3837	3837	3837	3837	3837

Table 7: Corporate social responsibility and operating cash flow, dividend payout

The table reports the relationships between corporate social responsibility and operating cash flow, dividend payout. Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

Variable	**Operating cash flow(CFO/TA)**		**Dividend payout**	
Intercept	-0.704	0.021	29.229***	33.703***
	(-0.200)	(0.430)	(3.710)	(4.160)
CSR	0.047*	0.002***	0.104***	0.166***
	(1.920)	(3.470)	(1.890)	(2.290)
*CSR*Group*		-0.001***		-0.314***
		(-2.380)		(-2.310)
Group	0.973*	0.029***	0.870	-3.617*
	(1.790)	(2.780)	(0.900)	(-1.670)
Size	-0.209	0.001	-0.530	-0.544*
	(-1.210)	(0.490)	(-1.620)	(-1.670)
Leverage	2.558	0.020	-18.161***	-18.379***
	(0.680)	(0.570)	(-6.560)	(-6.640)
ln(Firm Age)	-0.082	0.000	2.477***	2.408***
	(-0.240)	(0.010)	(3.590)	(3.490)
Board Ind(%)	-0.014	0.000	-0.076***	-0.081***
	(-0.840)	(-0.830)	(-2.210)	(-2.360)
Tobin Q	2.409***	2.109***	1.504***	1.520***
	(3.890)	(3.190)	(4.300)	(4.350)
Adjusted R^2	0.100	0.100	0.140	0.140
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
N	3837	3837	2385	2385

Table 8: Corporate social responsibility and operating firm performance

The table reports the relationships between corporate social responsibility and firm performace. Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

Variable	**ROA_{t+1}**	
Intercept	11.001**	9.279**
	(2.490)	(2.070)
CSR	0.075***	0.195***
	(3.140)	(3.370)
*CSR*Group*		-0.139**
		(-2.270)
Group	-0.438	1.528
	(-1.120)	(1.610)
Size	-0.282**	-0.275**
	(-2.320)	(-2.270)
Lev	-4.940***	-4.914***
	(-8.260)	(-8.220)
ln(Firm Age)	-0.117	-0.112
	(-0.410)	(-0.400)
Dividend Dummy	6.223***	6.218***
	(14.890)	(14.890)
Board Ind(%)	-0.012	-0.010
	(-0.860)	(-0.730)
Industry Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Adjusted R^2	0.276	0.267
N	3837	3837

Table 9: Corporate social responsibility: Financial crisis (2008-2009)

The table reports the effect of corporate social responsibility during financial crisis (2008-2009) . Table 1 reports the definition of variables. *t*-values are shown in parentheses.*, ** and *** indicate significant level at 10%, 5% and 1%, respectively.

Variable	Tobin Q	Cost of capital	Cost of debt	Cost of Equity
Intercept	0.954	8.215	3.941	8.410
	(1.400)	(5.330)	(1.830)	(6.740)
CSR	0.017***	-0.003	0.013	-0.025***
	(2.780)	(-0.300)	(0.940)	(-3.080)
Group	0.023	0.356***	0.132	0.305***
	(0.300)	(2.770)	(0.730)	(2.930)
Size	0.063***	0.096***	0.110***	0.337***
	(2.800)	(2.490)	(2.060)	(10.840)
Leverage	0.304***	-0.756***	1.126***	-0.126
	(3.530)	(-5.070)	(5.410)	(-1.050)
ln(Firm Age)	-0.041	0.116	0.091	-0.120
	(-0.740)	(1.240)	(0.700)	(-1.600)
Board Ind(%)	-0.003	0.001	0.005	0.000
	(-1.260)	(0.230)	(0.780)	(0.040)
Industry Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Adjusted R^2	0.236	0.285	0.164	0.353
N	858	858	858	858